

The Mixed Formulation and the Inf-Sup Condition: Why Coercivity Is Not Enough

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August 8, 2026

Abstract

In Lesson 2 I proved that the linear stationary micropolar system is well posed on the space of divergence free velocity fields. That result is correct but it cannot be used for computation, because there is no convenient finite element basis consisting of exactly divergence free functions. These notes therefore go back to the full problem, keep the pressure as an unknown, and impose incompressibility weakly. The price is that the resulting bilinear form is no longer coercive, so Lax–Milgram no longer applies. It has to be replaced by Brezzi’s theorem, which asks for two conditions instead of one: coercivity on the kernel, which is exactly what Lesson 2 already proved, and a new condition called the inf-sup condition. I set up all the spaces carefully, explain what every term in the inf-sup condition means and why it has the shape it does, and then work through five tasks establishing the structure of the mixed problem.

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1 Why we are changing the formulation

Lesson 2 worked on

$$\mathbf{V}_{\text{div}} = \{\mathbf{v} \in [H_0^1(\Omega)]^2 : \text{div } \mathbf{v} = 0\},$$

and the pressure disappeared from the weak form because the term $-\int_{\Omega} p \text{div } \mathbf{v} \, dx$ vanishes identically when $\mathbf{v}$ is divergence free. That gave a clean coercive problem and Lax–Milgram applied directly.

The difficulty is entirely computational. To discretise a problem posed on $\mathbf{V}_{\text{div}}$ we would need a finite element basis whose functions are *exactly* divergence free, that is, piecewise polynomial vector fields $\mathbf{v}_h$ with $\text{div } \mathbf{v}_h = 0$ pointwise. Such bases do exist, the Scott–Vogelius element being the best known example, but they require special meshes or high polynomial degree, and they are not what standard finite element software provides. If we take the ordinary vector valued P_2 basis, not a single one of its functions is divergence free.

So we go back to the full system, keep the pressure p as a genuine unknown, and impose $\text{div } \mathbf{u} = 0$ *weakly*, meaning as an equation tested against pressure functions rather than as a property built into the space. This is called the mixed formulation, and it is what every practical finite element code for incompressible flow actually solves.

The price is the subject of these notes. Once the space is enlarged, the bilinear form stops being coercive, and the theory has to change.

2 The spaces

Since the whole lesson is about which space each unknown lives in, I want to set all of them out carefully, with definitions, before anything else happens.

2.1 Recalling L^2 and H^1

Definition 2.1 (The space $L^2(\Omega)$). $L^2(\Omega)$ is the set of measurable functions $\phi : \Omega \rightarrow \mathbb{R}$ with

$$\int_{\Omega} |\phi|^2 \, dx < \infty,$$

where two functions are regarded as the same element if they agree outside a set of measure zero. It is equipped with the inner product and norm

$$(\phi, \psi) = \int_{\Omega} \phi \psi \, dx, \quad \|\phi\|_{L^2} = (\phi, \phi)^{1/2}.$$

The intuition, from Lesson 1, is that $\|\phi\|_{L^2}^2$ is a total squared magnitude. When ϕ is a velocity it is kinetic energy up to a constant, so membership of $L^2(\Omega)$ is the statement that the field has finite energy.

The identification of functions agreeing almost everywhere is needed for $\|\cdot\|_{L^2}$ to be a genuine norm rather than a seminorm. If we did not identify them, a function which is zero except at one point would have zero norm without being the zero function, and the axiom $\|\phi\|_{L^2} = 0 \Rightarrow \phi = 0$ would fail.

Definition 2.2 (The spaces $H^1(\Omega)$ and $H_0^1(\Omega)$). $H^1(\Omega)$ is the set of $\phi \in L^2(\Omega)$ whose weak first derivatives also lie in $L^2(\Omega)$, with norm

$$\|\phi\|_{H^1}^2 = \|\phi\|_{L^2}^2 + \|\nabla\phi\|_{L^2}^2.$$

The subspace $H_0^1(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in this norm, which for a Lipschitz domain coincides with the set of $H^1(\Omega)$ functions whose trace on $\partial\Omega$ vanishes.

Physically, $\|\nabla\phi\|_{L^2}^2$ measures how fast the field varies, and for a velocity field it is proportional to the rate of viscous energy dissipation. So $H^1(\Omega)$ is the space of fields with finite energy and finite dissipation rate, and $H_0^1(\Omega)$ imposes in addition the no slip condition at the walls.

2.2 The three spaces of the mixed problem

The mixed formulation has three unknowns and therefore three spaces.

Definition 2.3 (Velocity space).

$$\mathbf{V} = [H_0^1(\Omega)]^2 = \{\mathbf{v} = (v_1, v_2) : v_1, v_2 \in H_0^1(\Omega)\}, \quad \|\mathbf{v}\|_{H^1}^2 = \|v_1\|_{H^1}^2 + \|v_2\|_{H^1}^2.$$

Note carefully what has changed from Lesson 2. There we used $\mathbf{V}_{\text{div}}$, which was $\mathbf{V}$ together with the extra requirement $\text{div } \mathbf{v} = 0$. Now that requirement has been removed from the space. The velocity space is the plain product of two copies of $H_0^1(\Omega)$, with no constraint at all. The constraint has not been abandoned, it has been moved: it will reappear in Section 3 as an equation rather than as a property of the space.

Definition 2.4 (Microrotation space).

$$W = H_0^1(\Omega).$$

This is unchanged from Lesson 2. The microrotation is a scalar in two dimensions, it satisfies a homogeneous Dirichlet condition, and its equation is of reaction diffusion type, so $H_0^1(\Omega)$ is the natural space for exactly the reasons given in Lesson 1.

Definition 2.5 (Pressure space).

$$Q = L_0^2(\Omega) = \left\{ q \in L^2(\Omega) : \int_{\Omega} q \, dx = 0 \right\}, \quad \|q\|_Q = \|q\|_{L^2}.$$

This space is new, since in Lesson 2 there was no pressure unknown at all, so I want to say carefully both why the pressure lives in $L^2(\Omega)$ rather than $H^1(\Omega)$, and why we take the zero mean subspace.

Why L^2 and not H^1 . The rule from Lesson 1 is that the space is read off from the weak formulation by asking what regularity makes its integrals finite. In the mixed formulation the pressure appears only in the term

$$- \int_{\Omega} p \, \text{div } \mathbf{v} \, dx.$$

There is no derivative on p here. It was removed by integration by parts and transferred onto $\mathbf{v}$, which is exactly the trade we always make. For this integral to be finite we need only $p \in L^2(\Omega)$, since $\text{div } \mathbf{v} \in L^2(\Omega)$ whenever $\mathbf{v} \in \mathbf{V}$ and Cauchy–Schwarz then applies. Asking for $p \in H^1(\Omega)$ would be demanding more regularity than the formulation requires, and by the argument of Lesson 1 that is the wrong thing to do. So the pressure sits in a space with no derivatives at all, one step rougher than the velocity. This asymmetry between the velocity space and the pressure space is characteristic of mixed problems.

Why the zero mean condition. In the momentum equation the pressure appears only through ∇p . Hence if p solves the problem then so does $p + C$ for any constant C , because $\nabla(p + C) = \nabla p$. The physics determines pressure differences, not absolute pressure, so without some normalisation the solution is not unique.

The set of constants is a one dimensional subspace of $L^2(\Omega)$, and $L^2(\Omega)$ splits orthogonally as

$$L^2(\Omega) = L_0^2(\Omega) \oplus \{\text{constants}\}.$$

To see this, given $q \in L^2(\Omega)$ write

$$q = \underbrace{\left(q - \frac{1}{|\Omega|} \int_{\Omega} q \, dx \right)}_{\text{zero mean}} + \underbrace{\frac{1}{|\Omega|} \int_{\Omega} q \, dx}_{\text{constant}},$$

and check that the first bracket integrates to zero while the two pieces are orthogonal in $L^2(\Omega)$. Restricting to $L_0^2(\Omega)$ therefore amounts to choosing one representative from each family $\{p + C : C \in \mathbb{R}\}$, namely the one with zero average.

Remark 2.6 (What quotienting means, and where you have already seen it). To quotient a space by a subspace means to declare two elements equal whenever they differ by an element of that subspace, and then to work with the resulting families rather than with individual elements. Clock arithmetic is the familiar example: 14 and 2 are the same time because they differ by a multiple of 12.

Here the subspace is the constants, and p and $p + C$ are the same physical state. Rather than carrying families around we pick one representative from each, which is what Definition 2.5 does.

This is not an abstraction with no consequences. If the pressure space is left as all of $L^2(\Omega)$, so that the constants are still present, then the corresponding matrix in a finite element code has a nontrivial null space: the vector of nodal values representing a constant pressure is mapped to zero. The matrix is then singular, and a direct solver will either fail or return nonsense. The step of fixing the constant in code, whether by a Lagrange multiplier or by pinning one pressure degree of freedom, is the discrete counterpart of passing to $L_0^2(\Omega)$. So the abstract quotient and the singular matrix are the same phenomenon seen from two sides.

2.3 The product spaces

Following the packaging idea of Lesson 2, we group the unknowns. Write

$$\mathbf{V} \times W \quad \text{with} \quad \|(\mathbf{v}, z)\|_{\mathbf{V} \times W}^2 = \|\mathbf{v}\|_{H^1}^2 + \|z\|_{H^1}^2,$$

for the pair carrying the coercive part of the problem, and

$$X = (\mathbf{V} \times W) \times Q \quad \text{with} \quad \|(\mathbf{v}, z, q)\|_X^2 = \|\mathbf{v}\|_{H^1}^2 + \|z\|_{H^1}^2 + \|q\|_{L^2}^2,$$

for all three unknowns together. Each is a Hilbert space, being a finite product of Hilbert spaces with the sum of squares norm.

I will use $\mathbf{V} \times W$ throughout, and X only in Section 4, where the point is precisely that working on X does not succeed.

3 The mixed formulation

3.1 The strong form

The system, as in Lesson 2, is

$$-(\nu + \nu_r)\Delta \mathbf{u} + \nabla p = 2\nu_r \nabla^\perp w + \mathbf{f}, \quad (1)$$

$$\operatorname{div} \mathbf{u} = 0, \quad (2)$$

$$-(c_a + c_d)\Delta w + 4\nu_r w = 2\nu_r \operatorname{curl} \mathbf{u} + g, \quad (3)$$

with $\mathbf{u} = \mathbf{0}$ and $w = 0$ on $\partial\Omega$.

3.2 Task 3.A: deriving the mixed weak form

The derivation follows Lesson 2 exactly, with one change: we do not discard the pressure term, because the test functions are no longer divergence free.

Step 1: the momentum equation. Multiply (1) by $\mathbf{v} \in \mathbf{V}$ and integrate over Ω :

$$-(\nu + \nu_r) \int_{\Omega} (\Delta \mathbf{u}) \cdot \mathbf{v} \, dx + \int_{\Omega} \nabla p \cdot \mathbf{v} \, dx = 2\nu_r \int_{\Omega} \nabla^\perp w \cdot \mathbf{v} \, dx + \int_{\Omega} \mathbf{f} \cdot \mathbf{v} \, dx.$$

For the Laplacian term, apply Green's identity componentwise. The boundary terms vanish because $\mathbf{v} \in [H_0^1(\Omega)]^2$ has zero trace:

$$-(\nu + \nu_r) \int_{\Omega} (\Delta \mathbf{u}) \cdot \mathbf{v} \, dx = (\nu + \nu_r) (\nabla \mathbf{u}, \nabla \mathbf{v}).$$

For the pressure term, apply the integration by parts formula

$$\int_{\Omega} (\partial_i \phi) \psi \, dx = \int_{\partial\Omega} \phi \psi n_i \, ds - \int_{\Omega} \phi (\partial_i \psi) \, dx$$

with $\phi = p$ and $\psi = v_i$, once for each component. The boundary terms vanish because v_i has zero trace, so

$$\int_{\Omega} \nabla p \cdot \mathbf{v} \, dx = \sum_{i=1}^2 \int_{\Omega} (\partial_i p) v_i \, dx = - \sum_{i=1}^2 \int_{\Omega} p (\partial_i v_i) \, dx = - \int_{\Omega} p \operatorname{div} \mathbf{v} \, dx. \quad (4)$$

In Lesson 2 this was the end of the story, because $\operatorname{div} \mathbf{v} = 0$ made (4) vanish. Now $\mathbf{v}$ is an arbitrary element of $\mathbf{V}$, so the term survives. Define

$$b(\mathbf{v}, q) := - \int_{\Omega} q \operatorname{div} \mathbf{v} \, dx. \quad (5)$$

The momentum equation in weak form is then

$$(\nu + \nu_r) (\nabla \mathbf{u}, \nabla \mathbf{v}) - 2\nu_r (\nabla^\perp w, \mathbf{v}) + b(\mathbf{v}, p) = (\mathbf{f}, \mathbf{v}) \quad \text{for all } \mathbf{v} \in \mathbf{V}. \quad (6)$$

Step 2: the microrotation equation. Unchanged from Lesson 2. Multiply (3) by $z \in W$, integrate, apply Green's identity to the Laplacian term with the boundary term vanishing because z has zero trace, and leave the zeroth order term alone since it contains no derivative:

$$(c_a + c_d) (\nabla w, \nabla z) + 4\nu_r (w, z) - 2\nu_r (\operatorname{curl} \mathbf{u}, z) = (g, z) \quad \text{for all } z \in W. \quad (7)$$

Step 3: the incompressibility equation. This is the genuinely new step. In Lesson 2 the constraint was built into the space and never tested. Now we test it. Multiply (2) by $q \in Q$ and integrate:

$$\int_{\Omega} q \operatorname{div} \mathbf{u} \, dx = 0 \quad \text{for all } q \in Q,$$

which in the notation (5) is

$$b(\mathbf{u}, q) = 0 \quad \text{for all } q \in Q. \quad (8)$$

No integration by parts is needed, since there is nothing to move.

Step 4: adding the first two equations. As in Lesson 2, testing (6) against every $\mathbf{v}$ and (7) against every z is equivalent to testing their sum against every pair $(\mathbf{v}, z)$: choosing $z = 0$ recovers the first, choosing $\mathbf{v} = \mathbf{0}$ recovers the second. Writing a for the bilinear form of Lesson 2,

$$\begin{aligned} a((\mathbf{u}, w), (\mathbf{v}, z)) &= (\nu + \nu_r) (\nabla \mathbf{u}, \nabla \mathbf{v}) + (c_a + c_d) (\nabla w, \nabla z) + 4\nu_r (w, z) \\ &\quad - 2\nu_r (\nabla^\perp w, \mathbf{v}) - 2\nu_r (\operatorname{curl} \mathbf{u}, z), \end{aligned} \quad (9)$$

and $F(\mathbf{v}, z) = (\mathbf{f}, \mathbf{v}) + (g, z)$, we arrive at the mixed problem.

Definition 3.1 (Mixed weak formulation). Find $((\mathbf{u}, w), p) \in (\mathbf{V} \times W) \times Q$ such that

$$a((\mathbf{u}, w), (\mathbf{v}, z)) + b(\mathbf{v}, p) = F(\mathbf{v}, z) \quad \text{for all } (\mathbf{v}, z) \in \mathbf{V} \times W, \quad (10)$$

$$b(\mathbf{u}, q) = 0 \quad \text{for all } q \in Q. \quad (11)$$

Note the structure. The form a is exactly the one analysed in Lesson 2, with no change whatsoever; the only difference is that it is now posed on the larger space $\mathbf{V} \times W$ rather than $\mathbf{V}_{\operatorname{div}} \times W$. The new object is b , which appears twice, once carrying the pressure into the momentum equation and once expressing the constraint. A system with this shape, a coercive block plus a constraint block appearing symmetrically, is called a saddle point problem.

4 Task 3.B: the form is not coercive on X

The natural first attempt is to package all three unknowns into one bilinear form on X and apply Lax–Milgram as before. This section shows that it does not work, and identifies the reason.

Define, for $(\mathbf{u}, w, p)$ and $(\mathbf{v}, z, q)$ in X ,

$$\mathcal{A}((\mathbf{u}, w, p), (\mathbf{v}, z, q)) := a((\mathbf{u}, w), (\mathbf{v}, z)) + b(\mathbf{v}, p) - b(\mathbf{u}, q). \quad (12)$$

The minus sign on the last term is a convention, chosen so that the second equation of Definition 3.1 is recovered with the correct sign when we test with q ; any consistent choice would do.

Proposition 4.1 (No coercivity on X). *There is no constant $\alpha > 0$ with*

$$\mathcal{A}((\mathbf{u}, w, p), (\mathbf{u}, w, p)) \geq \alpha \|(\mathbf{u}, w, p)\|_X^2 \quad \text{for all } (\mathbf{u}, w, p) \in X.$$

Proof. First compute what happens on the diagonal. Setting $\mathbf{v} = \mathbf{u}$, $z = w$ and $q = p$ in (12),

$$\mathcal{A}((\mathbf{u}, w, p), (\mathbf{u}, w, p)) = a((\mathbf{u}, w), (\mathbf{u}, w)) + b(\mathbf{u}, p) - b(\mathbf{u}, p) = a((\mathbf{u}, w), (\mathbf{u}, w)).$$

The two occurrences of b cancel exactly. This is not an accident of the sign convention: whatever sign is chosen, the pressure enters $\mathcal{A}$ only through b , and on the diagonal both occurrences have the same arguments, so they cancel or double but in neither case produce anything involving $\|p\|_{L^2}^2$.

Now take the explicit choice

$$\mathbf{u} = \mathbf{0}, \quad w = 0, \quad p = q_0$$

where $q_0 \in Q$ is any nonzero element, for instance $q_0(x, y) = \cos(\pi x)$ on $\Omega = (0, 1)^2$, which has zero mean since $\int_0^1 \cos(\pi x) dx = 0$ and is not identically zero.

For this choice the left hand side is

$$\mathcal{A}((\mathbf{0}, 0, q_0), (\mathbf{0}, 0, q_0)) = a((\mathbf{0}, 0), (\mathbf{0}, 0)) = 0,$$

since a is bilinear and therefore vanishes when either argument is zero. The right hand side is

$$\alpha \|(\mathbf{0}, 0, q_0)\|_X^2 = \alpha (0 + 0 + \|q_0\|_{L^2}^2) = \alpha \|q_0\|_{L^2}^2 > 0,$$

because $q_0 \neq 0$. So the asserted inequality would read $0 \geq \alpha \|q_0\|_{L^2}^2 > 0$, which is false for every $\alpha > 0$. $\square$

Remark 4.2 (Why this is structural rather than a failure of technique). It is worth being clear that Proposition 4.1 is not saying that we were not clever enough. Coercivity on X would require the form to control $\|p\|_{L^2}^2$ from below, and the form simply contains no term in which the pressure is paired with itself. Compare with the velocity, which appears in $(\nu + \nu_r)(\nabla \mathbf{u}, \nabla \mathbf{v})$ and so produces $\|\nabla \mathbf{u}\|_{L^2}^2$ on the diagonal, or the microrotation, which appears in $4\nu_r(w, z)$ and produces $\|w\|_{L^2}^2$.

The reason is physical. Velocity and microrotation are diffusing fields, and the diffusion terms represent dissipation of their energy. The pressure is not a field of that kind at all. It is a Lagrange multiplier: a quantity whose only role is to enforce the constraint $\operatorname{div} \mathbf{u} = 0$, adjusting itself to whatever value is needed. It has no equation of its own, no diffusion, and no energy. A Lagrange multiplier can never appear coercively, in any problem, so no amount of manipulation will produce a $\|p\|_{L^2}^2$ term.

The conclusion is that Lax–Milgram is simply the wrong theorem here, and we need one designed for problems with constraints.

5 Brezzi's theorem

5.1 The kernel

The replacement theorem is stated in terms of the set of velocities that satisfy the constraint.

Definition 5.1 (Kernel of b).

$$Z = \{(\mathbf{v}, z) \in \mathbf{V} \times W : b(\mathbf{v}, q) = 0 \text{ for all } q \in Q\}.$$

5.2 Statement

Theorem 5.2 (Brezzi). *Let $\mathbf{V} \times W$ and Q be Hilbert spaces, let a be a bounded bilinear form on $(\mathbf{V} \times W)^2$ and b a bounded bilinear form on $(\mathbf{V} \times W) \times Q$. Suppose:*

(B1) Coercivity on the kernel. *There is $\alpha > 0$ with*

$$a((\mathbf{v}, z), (\mathbf{v}, z)) \geq \alpha \|(\mathbf{v}, z)\|_{\mathbf{V} \times W}^2 \quad \text{for all } (\mathbf{v}, z) \in Z.$$

(B2) The inf-sup condition. *There is $\beta > 0$ with*

$$\inf_{q \in Q \setminus \{0\}} \sup_{\mathbf{v} \in \mathbf{V} \setminus \{0\}} \frac{b(\mathbf{v}, q)}{\|\mathbf{v}\|_{H^1} \|q\|_{L^2}} \geq \beta.$$

Then for every bounded linear functional F the mixed problem of Definition 3.1 has exactly one solution $((\mathbf{u}, w), p)$, and it depends continuously on the data, with constants depending only on α , β and the boundedness constants of a and b .

I am quoting this theorem rather than proving it. The proof is not short and it belongs to the general theory of saddle point problems; Boffi, Brezzi and Fortin give it in full.

5.3 What has changed, and what has not

Two hypotheses now replace the single coercivity hypothesis of Lax–Milgram.

Condition (B1) is weaker than full coercivity, because it is only required on the kernel Z rather than on all of $\mathbf{V} \times W$. That is exactly what makes the theorem usable: we saw in Proposition 4.1 that coercivity on the whole space is hopeless, but on the kernel the constraint is satisfied and the situation is the one from Lesson 2.

Condition (B2) is entirely new and is the price of enlarging the space. It is what replaces the information we lost when we stopped building the constraint into the velocity space.

The important practical point is that Lesson 2 has already done the work for (B1). Task 3.C below shows that $Z = \mathbf{V}_{\text{div}} \times W$, which is precisely the space on which Lesson 2 established coercivity with

$$\alpha = \frac{\min(\nu, c_a + c_d)}{C_\Omega^2 + 1}.$$

So (B1) is already proved, and the whole of the remaining difficulty is (B2).

6 Reading the inf-sup condition

The inf-sup condition is the least transparent object in this subject on first meeting, so this section takes it apart term by term.

6.1 Unpacking the notation

The expression is

$$\inf_{q \neq 0} \sup_{\mathbf{v} \neq \mathbf{0}} \frac{b(\mathbf{v}, q)}{\|\mathbf{v}\|_{H^1} \|q\|_{L^2}} \geq \beta.$$

Read from the inside out. For a fixed q , the inner supremum asks: over all possible velocity test functions $\mathbf{v}$, how large can the quantity $b(\mathbf{v}, q)$ be made, once we normalise by the sizes of $\mathbf{v}$ and q so that the answer does not depend on scaling? Call this number $S(q)$.

The outer infimum then asks: as q ranges over all nonzero pressures, how small can $S(q)$ get? The condition demands that it never drops below the fixed positive number β .

Equivalently, and this form is usually easier to work with: for every nonzero $q \in Q$ there exists $\mathbf{v} \in \mathbf{V}$ with

$$b(\mathbf{v}, q) \geq \beta \|\mathbf{v}\|_{H^1} \|q\|_{L^2}. \quad (13)$$

6.2 What it says in words

Since $b(\mathbf{v}, q) = -\int_\Omega q \operatorname{div} \mathbf{v} \, dx$, the condition says that *every pressure must be visible to some velocity*. For each candidate pressure q , there has to be a velocity field whose divergence pairs with q in a nontrivial way.

Suppose instead that some nonzero q_0 had $b(\mathbf{v}, q_0) = 0$ for every $\mathbf{v} \in \mathbf{V}$. Then q_0 would not appear anywhere in equation (10), since that equation only ever sees the pressure through $b(\mathbf{v}, \cdot)$. Consequently, if $((\mathbf{u}, w), p)$ were a solution, so would $((\mathbf{u}, w), p + q_0)$ be, and uniqueness would fail. The inf-sup condition is what rules this out.

6.3 Why a uniform constant is needed

It would be a weaker requirement to ask merely that for each nonzero q there exists some $\mathbf{v}$ with $b(\mathbf{v}, q) \neq 0$. That is a statement about each q separately. The inf-sup condition asks for more: a single constant β that works for all q simultaneously. This is what the word *uniform* means here.

The reason the stronger version is needed is as follows. Suppose the weaker statement held but no uniform β existed. Then there would be a sequence q_n of pressures, each visible to some velocity, but with

$$S(q_n) \longrightarrow 0.$$

Each q_n is detected by the equations, but progressively more feebly. Now, Brezzi's theorem produces a bound on the solution of the form

$$\|p\|_{L^2} \leq \frac{C}{\beta} (\text{data}),$$

so as β degenerates this bound blows up. The map taking the data F to the solution p is called the *solution operator*, and to say it is *unbounded* is to say that no such finite constant exists: arbitrarily small data could produce arbitrarily large pressure. That is a failure of stability, and a problem with an unbounded solution operator is not well posed in any useful sense, even if a solution technically exists and is unique.

This has a very concrete numerical consequence, which is the subject of Lesson 4. In the discrete setting the constant β_h for the finite element spaces may degenerate as the mesh is refined, and when it does the computed pressure develops large spurious oscillations even though the velocity may look perfectly reasonable.

6.4 Why the pressure space had to be $L_0^2(\Omega)$

Task 3.E below shows that the inf-sup condition fails outright if the pressure space is taken to be all of $L^2(\Omega)$, because constant pressures are invisible to every velocity in $[H_0^1(\Omega)]^2$. Passing to $L_0^2(\Omega)$ removes exactly those invisible elements, and by the discussion in Remark 2.6 this is the same step as fixing the singular matrix in a computation. The abstract requirement and the practical fix are two views of one fact.

6.5 Status of the continuous inf-sup condition

For the specific form $b(\mathbf{v}, q) = -\int_{\Omega} q \operatorname{div} \mathbf{v} \, dx$ on $[H_0^1(\Omega)]^2 \times L_0^2(\Omega)$, condition (B2) does hold on any bounded Lipschitz domain. It is equivalent to the statement that the divergence operator has a bounded right inverse: given $q \in L_0^2(\Omega)$, there exists $\mathbf{v} \in [H_0^1(\Omega)]^2$ with

$$\operatorname{div} \mathbf{v} = q \quad \text{and} \quad \|\mathbf{v}\|_{H^1} \leq C \|q\|_{L^2}.$$

Constructing such a $\mathbf{v}$ is a substantial theorem, associated with the names of Bogovskii and of Nečas. I am citing it, not proving it. Deciding what to prove and what to quote is itself part of doing this properly, and this one is firmly on the quoting side.

7 Task 3.C: identifying the kernel

Proposition 7.1. *With Z as in Definition 5.1,*

$$Z = \mathbf{V}_{\operatorname{div}} \times W.$$

Before the proof, note where the difficulty lies. We want to conclude from

$$\int_{\Omega} q \operatorname{div} \mathbf{v} \, dx = 0 \quad \text{for all } q \in Q$$

that $\operatorname{div} \mathbf{v} = 0$. The obvious move is to choose $q = \operatorname{div} \mathbf{v}$, which would give $\|\operatorname{div} \mathbf{v}\|_{L^2}^2 = 0$ immediately. But that choice is only legitimate if $\operatorname{div} \mathbf{v}$ actually belongs to $Q = L_0^2(\Omega)$, and membership of $L_0^2(\Omega)$ requires zero mean, which is not automatic. So the first step of the proof must be to check that $\operatorname{div} \mathbf{v}$ does have zero mean.

Proof. The microrotation component plays no role, since z does not appear in b at all. So $(\mathbf{v}, z) \in Z$ if and only if $z \in W$ and $b(\mathbf{v}, q) = 0$ for all $q \in Q$. It therefore suffices to show that the latter condition is equivalent to $\operatorname{div} \mathbf{v} = 0$.

Step 1: $\operatorname{div} \mathbf{v}$ has zero mean. Let $\mathbf{v} \in \mathbf{V} = [H_0^1(\Omega)]^2$. By the divergence theorem,

$$\int_{\Omega} \operatorname{div} \mathbf{v} \, dx = \int_{\partial\Omega} \mathbf{v} \cdot \mathbf{n} \, ds.$$

Since $\mathbf{v}$ has zero trace on $\partial\Omega$, the integrand on the right vanishes identically, so

$$\int_{\Omega} \operatorname{div} \mathbf{v} \, dx = 0. \tag{14}$$

Also $\operatorname{div} \mathbf{v} \in L^2(\Omega)$, because $\mathbf{v} \in [H^1(\Omega)]^2$ means each $\partial_i v_i \in L^2(\Omega)$ and a sum of $L^2(\Omega)$ functions is in $L^2(\Omega)$. Combining these two facts,

$$\operatorname{div} \mathbf{v} \in L_0^2(\Omega) = Q.$$

So $\operatorname{div} \mathbf{v}$ is an admissible choice of test function, which is what we needed to establish.

Step 2: the forward implication. Suppose $b(\mathbf{v}, q) = 0$ for all $q \in Q$. By Step 1 we may take $q = \operatorname{div} \mathbf{v}$, giving

$$0 = b(\mathbf{v}, \operatorname{div} \mathbf{v}) = - \int_{\Omega} (\operatorname{div} \mathbf{v})(\operatorname{div} \mathbf{v}) \, dx = - \|\operatorname{div} \mathbf{v}\|_{L^2}^2.$$

Hence $\|\operatorname{div} \mathbf{v}\|_{L^2} = 0$, and since $\|\cdot\|_{L^2}$ is a norm this forces $\operatorname{div} \mathbf{v} = 0$ as an element of $L^2(\Omega)$, that is, almost everywhere in Ω . Therefore $\mathbf{v} \in \mathbf{V}_{\operatorname{div}}$.

Step 3: the reverse implication. Conversely, if $\mathbf{v} \in \mathbf{V}_{\operatorname{div}}$ then $\operatorname{div} \mathbf{v} = 0$, so for every $q \in Q$

$$b(\mathbf{v}, q) = - \int_{\Omega} q \cdot 0 \, dx = 0,$$

and $(\mathbf{v}, z) \in Z$ for any $z \in W$.

Combining Steps 2 and 3 gives $Z = \mathbf{V}_{\operatorname{div}} \times W$. □

Remark 7.2. This proposition is what connects the two lessons. Condition (B1) of Brezzi's theorem asks for coercivity of a on Z , and Proposition 7.1 identifies Z as $\mathbf{V}_{\operatorname{div}} \times W$, which is exactly the space on which Lesson 2 proved coercivity. So (B1) holds with $\alpha = \min(\nu, c_a + c_d)/(C_{\Omega}^2 + 1)$, and nothing from Lesson 2 is wasted. The whole of the analysis done there transfers to the mixed setting without a single change.

8 Task 3.D: boundedness of b

Brezzi's theorem requires b to be bounded, so we check it. The proof needs one small inequality which is worth isolating.

Lemma 8.1. *For $\mathbf{v} \in [H^1(\Omega)]^2$,*

$$\|\operatorname{div} \mathbf{v}\|_{L^2} \leq \sqrt{2} \|\nabla \mathbf{v}\|_{L^2}.$$

Proof. Pointwise, $\operatorname{div} \mathbf{v} = \partial_1 v_1 + \partial_2 v_2$ is a sum of two numbers. Apply the Cauchy–Schwarz inequality in $\mathbb{R}^2$ to the vectors $(1, 1)$ and $(\partial_1 v_1, \partial_2 v_2)$:

$$|\partial_1 v_1 + \partial_2 v_2| \leq \sqrt{1^2 + 1^2} \sqrt{(\partial_1 v_1)^2 + (\partial_2 v_2)^2} = \sqrt{2} \sqrt{(\partial_1 v_1)^2 + (\partial_2 v_2)^2}.$$

Squaring and integrating over Ω ,

$$\|\operatorname{div} \mathbf{v}\|_{L^2}^2 \leq 2 \int_{\Omega} [(\partial_1 v_1)^2 + (\partial_2 v_2)^2] \, dx \leq 2 \int_{\Omega} [(\partial_1 v_1)^2 + (\partial_2 v_1)^2 + (\partial_1 v_2)^2 + (\partial_2 v_2)^2] \, dx,$$

the last step being the discard of two non-negative terms, or rather the addition of them. The final integral is exactly $\|\nabla \mathbf{v}\|_{L^2}^2$, so $\|\operatorname{div} \mathbf{v}\|_{L^2}^2 \leq 2 \|\nabla \mathbf{v}\|_{L^2}^2$, and taking square roots gives the claim. $\square$

Remark 8.2. It is worth noting why Lemma ?? of Lesson 2, which stated $\|\nabla \mathbf{v}\|_{L^2}^2 = \|\operatorname{curl} \mathbf{v}\|_{L^2}^2 + \|\operatorname{div} \mathbf{v}\|_{L^2}^2$ and would give the sharper bound $\|\operatorname{div} \mathbf{v}\|_{L^2} \leq \|\nabla \mathbf{v}\|_{L^2}$, is not used here. That identity required $\mathbf{v} \in [H_0^1(\Omega)]^2$, because its proof used integration by parts twice and needed the boundary terms to vanish. Lemma 8.1 makes no use of boundary conditions and holds on all of $[H^1(\Omega)]^2$, at the cost of the constant $\sqrt{2}$. Since we will eventually want boundedness of b in settings where the boundary conditions may differ, the more robust version is the right one to record.

Proposition 8.3 (Boundedness of b). *For all $\mathbf{v} \in \mathbf{V}$ and $q \in Q$,*

$$|b(\mathbf{v}, q)| \leq \sqrt{2} \|\mathbf{v}\|_{H^1} \|q\|_{L^2}.$$

Proof. By the definition (5) and Cauchy–Schwarz in $L^2(\Omega)$,

$$|b(\mathbf{v}, q)| = \left| \int_{\Omega} q \operatorname{div} \mathbf{v} \, dx \right| \leq \|q\|_{L^2} \|\operatorname{div} \mathbf{v}\|_{L^2}.$$

By Lemma 8.1, $\|\operatorname{div} \mathbf{v}\|_{L^2} \leq \sqrt{2} \|\nabla \mathbf{v}\|_{L^2}$. Finally $\|\nabla \mathbf{v}\|_{L^2} \leq \|\mathbf{v}\|_{H^1}$, since $\|\mathbf{v}\|_{H^1}^2 = \|\mathbf{v}\|_{L^2}^2 + \|\nabla \mathbf{v}\|_{L^2}^2$ and $\|\mathbf{v}\|_{L^2}^2 \geq 0$, so discarding a non-negative term gives the inequality. Chaining the three estimates,

$$|b(\mathbf{v}, q)| \leq \|q\|_{L^2} \sqrt{2} \|\nabla \mathbf{v}\|_{L^2} \leq \sqrt{2} \|\mathbf{v}\|_{H^1} \|q\|_{L^2}. \quad \square$$

9 Task 3.E: failure of inf-sup on $L^2(\Omega)$

Proposition 9.1. *Let the pressure space be taken as all of $L^2(\Omega)$ rather than $L_0^2(\Omega)$. Then the inf-sup condition fails: there is no $\beta > 0$ with*

$$\sup_{\mathbf{v} \in \mathbf{V} \setminus \{0\}} \frac{b(\mathbf{v}, q)}{\|\mathbf{v}\|_{H^1} \|q\|_{L^2}} \geq \beta \quad \text{for all } q \in L^2(\Omega) \setminus \{0\}.$$

Proof. Take $q \equiv 1$, which is a nonzero element of $L^2(\Omega)$ with $\|1\|_{L^2} = |\Omega|^{1/2} > 0$. For any $\mathbf{v} \in \mathbf{V} = [H_0^1(\Omega)]^2$,

$$b(\mathbf{v}, 1) = - \int_{\Omega} 1 \cdot \operatorname{div} \mathbf{v} \, dx = - \int_{\Omega} \operatorname{div} \mathbf{v} \, dx.$$

By the divergence theorem,

$$\int_{\Omega} \operatorname{div} \mathbf{v} \, dx = \int_{\partial\Omega} \mathbf{v} \cdot \mathbf{n} \, ds = 0,$$

since $\mathbf{v}$ has zero trace on $\partial\Omega$. Hence $b(\mathbf{v}, 1) = 0$ for every $\mathbf{v} \in \mathbf{V}$, and therefore

$$\sup_{\mathbf{v} \neq \mathbf{0}} \frac{b(\mathbf{v}, 1)}{\|\mathbf{v}\|_{H^1} \|1\|_{L^2}} = 0.$$

Since $\|1\|_{L^2} > 0$, no positive β can bound this supremum from below. $\square$

Remark 9.2. This is the same calculation as Step 1 of Proposition 7.1, seen from the other side. There it showed that $\operatorname{div} \mathbf{v}$ automatically has zero mean, which was what made the kernel argument work. Here the same fact shows that constant pressures are orthogonal to the range of the divergence operator, and so are invisible to the formulation.

The two are aspects of one statement: the divergence operator maps $[H_0^1(\Omega)]^2$ into $L_0^2(\Omega)$ and not onto anything larger. Restricting the pressure space to $L_0^2(\Omega)$ matches the pressure space to the range of the operator, which is exactly what the inf-sup condition requires, and it is also what removes the singular mode from the discrete matrix as described in Remark 2.6.

10 Summary and what comes next

10.1 Where the mixed problem stands

Collecting the results of this lesson against the hypotheses of Theorem 5.2:

Requirement	Status	Reference
a bounded on $\mathbf{V} \times W$	proved	Lesson 2, Theorem on boundedness
b bounded	proved	Proposition 8.3
Kernel identified as $\mathbf{V}_{\operatorname{div}} \times W$	proved	Proposition 7.1
(B1) coercivity on the kernel	proved	Lesson 2, coercivity theorem
(B2) inf-sup condition	quoted	Bogovskii, Nečas

So the continuous mixed problem is well posed, and every ingredient except the inf-sup condition itself was established by our own work.

10.2 The shape of the argument, in one paragraph

Enlarging the velocity space from $\mathbf{V}_{\operatorname{div}}$ to $\mathbf{V}$ made the problem discretisable but destroyed coercivity, because the pressure entered as a Lagrange multiplier and multipliers never appear coercively. Brezzi's theorem repairs this by weakening the coercivity requirement to hold only on the kernel of the constraint, where the original Lesson 2 analysis applies verbatim, and adding in exchange a new condition ensuring that no pressure is invisible to the velocity space. The zero mean normalisation of the pressure space is not a convention but a necessity, since without it the constants form exactly such an invisible set.

10.3 What comes next

Everything in this lesson concerned the continuous problem. The next lesson concerns the discrete one, and the central point is that condition (B2) does *not* pass automatically to finite element subspaces. Choosing $\mathbf{V}_h \subset \mathbf{V}$ and $Q_h \subset Q$ gives a discrete inf-sup constant β_h which may be small, or may degenerate as the mesh is refined, even though the continuous β is perfectly good.

This is the reason that equal order elements such as P_1/P_1 fail for Stokes, producing the characteristic checkerboard pressure oscillation, and the reason the Taylor–Hood pair P_2/P_1 is used instead. That is Lesson 4.